

Where Does the Hedging Error Go? A Greek-by-Greek Decomposition for At-the-Money 0DTE Options under Heston

Harshith Ghanta

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Abstract

Zero Days to Expiration Options (0DTE) have increased in popularity since CBOE released daily SPY expirations in 2022. However, they are difficult to hedge, and the sources of hedging error are not well understood. This paper takes the Heston model as a starting point and uses an industry-standard Greek-by-Greek decomposition to measure how much each Greek moves the profit and loss of an at-the-money delta hedge rebalanced at five-minute intervals, with the variance state read from Cboe's VIX1D index, which is horizon-matched to a 0DTE contract. Shares are reported two ways: as a signed decomposition of the variance of the realized hedged P&L, which is the primary measure, and as gross absolute attribution shares computed from model Greeks, which measure the size of each term's contribution rather than a split of the realized error. On the variance measure Gamma carries 64.6% and Vega 31.5%; the two together account for almost all of the variance of the hedged P&L, leaving 5.9% unexplained, and every other term sits within sampling error of zero. The two largest terms correlate at $\rho = +0.27$ across days, with a 95% interval that includes zero, so there is no evidence that they cancel within a day. Gamma dominates, absorbing both diffusion convexity and any jump risk the model cannot identify; a bipower-variation test flags 12 of the 57 days as containing jumps, and Gamma's share is unchanged on the jump-free subsample. On the days when VIX1D travels furthest intraday, Vega's gross share rises from 19.7% to 42.8%. The hedged P&L drifts negative but not significantly so, at 1.9 Newey–West standard errors of the mean; the drift that does appear term by term is a property of the variance proxy rather than a risk premium, since the model's variance exceeds realized variance on 51 of 57 days while VIX1D itself rises mechanically through the session on 55 of 57.

1 Introduction

This paper investigates the Greeks as sources of discrete hedging error for at-the-money 0DTE options under Heston. It takes the standard compensated-Greek decomposition of a delta-hedged option's profit and loss and applies it to at-the-money 0DTE calls, hedged at five-minute frequency on real SPY data with the variance state taken from the VIX1D index. The contribution is a term-by-term measurement of how much each Greek moves the hedged profit and loss, and how that ranking shifts with the volatility regime. Two measures are reported and they answer different questions: a signed decomposition of the variance of the realized hedged P&L, which is the primary measure, and a gross absolute attribution share, which records how much a term moves without regard to whether that movement survives into the realized error. The distinction is developed in Section 3.1. Gamma dominates in every regime; what the volatility regime changes is Vega, which becomes a material source of variance in the hedged P&L on the days VIX1D travels furthest, and which also acquires a drift that Section 3.1 attributes to the proxy rather than to the market. The endogeneity of that sort is discussed alongside the result.

1.1 Black–Scholes Model and Greeks

The Black–Scholes model, introduced by Fischer Black and Myron Scholes (1973) [1] and Robert Merton (1973) [2], provides a framework for pricing European options in a closed form that Heston expands on. The issue with Black–Scholes is that it assumes constant volatility, which is not the case in real markets, especially for 0DTE options. The Heston model, explained below, allows for stochastic volatility. Black–Scholes also introduced the concept of Greeks, which are sensitivities of the option price to various parameters. They are listed below:

- Delta (Δ): Sensitivity of the option price to changes in the underlying asset price.
- Gamma (Γ): Sensitivity of Delta to changes in the underlying asset price.
- Vega (ν): Sensitivity of the option price to changes in volatility.
- Theta (Θ): Sensitivity of the option price to the passage of time.
- Charm ($\frac{\partial \Delta}{\partial t}$): Sensitivity of Delta to the passage of time.

- Vanna ($\frac{\partial \Delta}{\partial \sigma}$): Sensitivity of Delta to changes in volatility.
- Volga, also called Vomma ($\frac{\partial \nu}{\partial \sigma}$): Sensitivity of Vega to changes in volatility.
- Veta ($\frac{\partial \nu}{\partial t}$): Sensitivity of Vega to the passage of time.
- Time convexity ($\frac{\partial^2 V}{\partial t^2}$): Second-order sensitivity of the option price to the passage of time.

1.2 Related Literature

The decomposition in this paper relies heavily on various prior works. The Heston model [3] is used as the underlying model for the option pricing and hedging found later in the paper. Along with that, the Euler–Maruyama scheme [18], in the standard form given by Kloeden and Platen [7], is used to simulate the stochastic differential equations used in the implementation of the paper. The Greek decomposition follows Bergomi [12]. Four of the seven terms used here are his, the Gamma, Volga, Vanna, and Vega terms, as is the compensator method that subtracts from each term the drift it earns under the model, so that what is left is the break-even quantity. The three remaining terms, Charm, Veta, and time convexity, are of order Δt and are the ODTE-specific addition: they are negligible at the maturities Bergomi treats, but the expansion here is carried to second order across a single session, where they do not vanish. Specifically, his “break-even” logic is used to attribute the hedging error to each Greek. Bergomi’s logic is the implied vs. realized volatility, where dominating volatility leads to price dominating the true price, and the Gamma variance drives the hedging error. This implied-versus-realized logic was formalized by El Karoui, Jeanblanc-Picqué, and Shreve (1998) [5], who show that the P&L of a delta hedge run at the wrong volatility is the dollar-Gamma-weighted spread between the hedging and realized variances. Bakshi and Kapadia (2003) [10] take that spread to the data, running delta-hedged S&P 500 option positions over horizons of days to months and finding their gains systematically negative, which they read as evidence of a negative market volatility risk premium. The single-session analogue is measured in Section 3.1 and does not reproduce it: the mean daily hedged P&L is negative but statistically insignificant, and the drifts that do appear term by term are traced there to the variance proxy rather than to a premium.

The result that discrete-hedging error scales as $\frac{1}{\sqrt{N}}$, where N is the number of rebalances, originates with Boyle and Emanuel (1980) [6], who derive

the variance of the hedging error of a discretely adjusted option position. Derman and Kamal [9] give the practitioner treatment of that result and the closest antecedent to this paper: a guide to hedging errors when one cannot hedge continuously, written for a trading desk rather than for a journal. Bertsimas–Kogan–Lo [11] approach the same problem independently and from the continuous-time side, finding that discrete tracking error grows with the granularity constant, g , which is built from accumulated dollar gamma. The formal contribution of Bertsimas et al. is the expression $\frac{g}{\sqrt{N}}$; they also show that the kink in the call payoff does not break this convergence rate, but rather inflates g . Dim, Eraker, and Vilkov (2023) [15] study how ODTE order imbalances and the resulting dealer gamma exposure propagate into market volatility. That channel is microstructural; the attribution here is at the contract level. No prior paper gives a term-by-term hedging-error attribution for ODTE options under Heston.

1.3 Choices and Rationale

The Heston model was chosen for its stochastic volatility and easily implementable characteristic function. Also, the Heston parameters are round, arbitrarily chosen values rather than calibrated estimates, except for η (see the Methodology section for further information). Their specific combination violates the Feller condition, so the variance process can reach zero; the simulation therefore uses a full-truncation Euler–Maruyama scheme [8], which floors the variance at zero at each step.

1.4 Heston Model Derivation

The derivation here follows both Gatheral (2006) [4] and Heston (1993) [3]. The derivation starts with the Heston model’s stochastic differential equations for the underlying asset price S and its variance v :

$$dS = \mu S dt + \sqrt{v} S dX, \quad dv = \kappa(\eta - v) dt + \epsilon \sqrt{v} dW, \quad dX dW = \rho dt,$$

There are two sources of randomness here, the price shock dX and the variance shock dW , so the hedge takes two instruments: the stock, and a second option V_1 with its own exposure to the variance. Take the portfolio, Π , with the option V by shorting ΔS , which is the dollar value of the stock shorted. Short Δ_1 shares of V_1 as well:

$$\Pi = V - \Delta S - \Delta_1 V_1.$$

Applying Itô's lemma to V gives its change over an instant:

$$dV = \left(\frac{\partial V}{\partial t} + \frac{1}{2}vS^2 \frac{\partial^2 V}{\partial S^2} + \rho\epsilon vS \frac{\partial^2 V}{\partial S \partial v} + \frac{1}{2}\epsilon^2 v \frac{\partial^2 V}{\partial v^2} \right) dt + \frac{\partial V}{\partial S} dS + \frac{\partial V}{\partial v} dv,$$

Apply the same expansion to V_1 , and substitute both into $d\Pi$ to obtain the following equation:

$$\begin{aligned} d\Pi = & \left(\frac{\partial V}{\partial t} + \frac{1}{2}vS^2 \frac{\partial^2 V}{\partial S^2} + \rho\epsilon vS \frac{\partial^2 V}{\partial S \partial v} + \frac{1}{2}\epsilon^2 v \frac{\partial^2 V}{\partial v^2} \right) dt \\ & - \Delta_1 \left(\frac{\partial V_1}{\partial t} + \frac{1}{2}vS^2 \frac{\partial^2 V_1}{\partial S^2} + \rho\epsilon vS \frac{\partial^2 V_1}{\partial S \partial v} + \frac{1}{2}\epsilon^2 v \frac{\partial^2 V_1}{\partial v^2} \right) dt \\ & + \left(\frac{\partial V}{\partial S} - \Delta_1 \frac{\partial V_1}{\partial S} - \Delta \right) dS + \left(\frac{\partial V}{\partial v} - \Delta_1 \frac{\partial V_1}{\partial v} \right) dv. \end{aligned}$$

The last line shows how to remove the risk: set Δ_1 to cancel the dv exposure, and Δ to cancel the dS exposure,

$$\Delta_1 = \frac{\partial V / \partial v}{\partial V_1 / \partial v}, \quad \Delta = \frac{\partial V}{\partial S} - \Delta_1 \frac{\partial V_1}{\partial S},$$

leaving a riskless portfolio. No-arbitrage then forces Π to earn the risk-free rate, $d\Pi = r\Pi dt$. Because the same argument holds for any option, the terms involving V and the terms involving V_1 must each equal a common function of (S, v, t) ; following Heston, that function is written as $-\nu_Q$:

$$\begin{aligned} \frac{1}{\partial V / \partial v} \left(\frac{\partial V}{\partial t} + \frac{1}{2}vS^2 \frac{\partial^2 V}{\partial S^2} + \rho\epsilon vS \frac{\partial^2 V}{\partial S \partial v} \right. \\ \left. + \frac{1}{2}\epsilon^2 v \frac{\partial^2 V}{\partial v^2} + rS \frac{\partial V}{\partial S} - rV \right) = -\nu_Q, \end{aligned}$$

where

$$\nu_Q \equiv \kappa(\eta - v) - \lambda v$$

is the risk-neutral drift of the variance, with λ the market price of volatility risk. Rearranging gives the Heston pricing PDE:

$$\begin{aligned} \frac{\partial V}{\partial t} + \frac{1}{2}vS^2 \frac{\partial^2 V}{\partial S^2} + \rho\epsilon vS \frac{\partial^2 V}{\partial S \partial v} + \frac{1}{2}\epsilon^2 v \frac{\partial^2 V}{\partial v^2} \\ + rS \frac{\partial V}{\partial S} + [\kappa(\eta - v) - \lambda v] \frac{\partial V}{\partial v} - rV = 0. \end{aligned}$$

2 The Error Model

2.1 Derivation

The Error model is derived from the Heston model; the two coincide in the continuous limit and diverge as the hedge becomes discrete. The Heston model assumes continuous hedging; therefore, $\Delta t \rightarrow 0$. However, in practice, hedging is done at discrete intervals, which introduces hedging error. To decompose said error, the following model is proposed based on Bergomi's compensated-Greek decomposition, adding the discrete-time terms (charm, veta, and time convexity) that arise from expanding the profit and loss to second order in Δt [12]:

$$\Delta \Pi = \Delta V - \frac{\partial V}{\partial S} \Delta S - r \left(V - \frac{\partial V}{\partial S} S \right) \Delta t.$$

$$\begin{aligned} \Delta \Pi = & \frac{\partial V}{\partial t} \Delta t + \frac{\partial V}{\partial S} \Delta S + \frac{\partial V}{\partial v} \Delta v \\ & + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} (\Delta S)^2 + \frac{1}{2} \frac{\partial^2 V}{\partial v^2} (\Delta v)^2 + \frac{\partial^2 V}{\partial S \partial v} \Delta S \Delta v \\ & + \frac{\partial^2 V}{\partial S \partial t} \Delta S \Delta t + \frac{\partial^2 V}{\partial v \partial t} \Delta v \Delta t + \frac{1}{2} \frac{\partial^2 V}{\partial t^2} (\Delta t)^2 \\ & - \frac{\partial V}{\partial S} \Delta S - r \left(V - \frac{\partial V}{\partial S} S \right) \Delta t. \end{aligned}$$

$$\frac{\partial V}{\partial t} = rV - rS \frac{\partial V}{\partial S} - \frac{1}{2} v S^2 \frac{\partial^2 V}{\partial S^2} - \rho \epsilon v S \frac{\partial^2 V}{\partial S \partial v} - \frac{1}{2} \epsilon^2 v \frac{\partial^2 V}{\partial v^2} - \nu_Q \frac{\partial V}{\partial v},$$

$$\begin{aligned} \Delta \Pi = & \frac{1}{2} \frac{\partial^2 V}{\partial S^2} [(\Delta S)^2 - v S^2 \Delta t] + \frac{1}{2} \frac{\partial^2 V}{\partial v^2} [(\Delta v)^2 - \epsilon^2 v \Delta t] \\ & + \frac{\partial^2 V}{\partial S \partial v} [\Delta S \Delta v - \rho \epsilon v S \Delta t] + \frac{\partial V}{\partial v} [\Delta v - \nu_Q \Delta t] \\ & + \frac{\partial^2 V}{\partial S \partial t} \Delta S \Delta t + \frac{\partial^2 V}{\partial v \partial t} \Delta v \Delta t + \frac{1}{2} \frac{\partial^2 V}{\partial t^2} (\Delta t)^2. \end{aligned}$$

Not every term above is discretization error. Summed over a day, the Gamma, Vanna, and Volga compensators vanish as the hedging interval shrinks, since realized quadratic variation converges to the model's $\int v S^2 dt$ and likewise for the cross and variance brackets, while the charm, veta, and time-convexity terms are of order Δt or smaller. Those six terms measure finite-mesh error: the price of rebalancing every five minutes instead

of continuously. The Vega term behaves otherwise. Its sum converges to $\int \frac{\partial V}{\partial v} \epsilon \sqrt{v} dW$, a random quantity that survives any rebalancing frequency. That is the variance risk a delta-only hedge leaves open, since the hedge holds no second option. The Heston derivation above removed this risk with the hedging option V_1 ; the experiment below deliberately holds only the stock, as most 0DTE hedgers do, so the Vega term tracks unhedged risk rather than finite-mesh error.

2.2 Methodology

The Heston parameters were not calibrated to data: apart from η , they are round, arbitrarily chosen values. η was set from the VIX1D-implied variance in the sample. Given that information, the results in the next section are not to be taken as absolute, but rather as a proof of concept. The parameters used are as follows:

- $\kappa = 3.0$ (rate of mean reversion)
- $\eta = 0.0160$ (long-run variance; the sample average of $(\text{VIX1D}/100)^2$ across the 57 days)
- $\epsilon = 1.0$ (volatility of volatility)
- $\rho = -0.7$ (correlation between asset and variance)
- $\lambda = 0$ (market price of volatility risk)
- $r = 0.045$ (risk-free rate, used in the financing term of the hedged P&L)

Two of these are uncalibrated free choices worth probing, κ and ϵ , and Section 3.1 reports a sweep over both. The mean-reversion rate κ is expected to be close to inert on this horizon: it enters only through $\nu_Q \Delta t = \kappa(\eta - v) \Delta t$, which for $\kappa = 3$ and a five-minute step is of order 10^{-6} , against a typical per-bar Δv two orders of magnitude larger. The vol-of-vol ϵ has no such protection, since it scales the Volga compensator as $\epsilon^2 v \Delta t$ and the Vanna compensator as $\rho \epsilon v S \Delta t$. The baseline is set at $\epsilon = 1.0$ rather than at the lower $\epsilon = 0.5$ deliberately. Calibrated short-horizon index vol-of-vol is commonly well above 0.5, and the sweep in Section 3.1 shows that the low value is the one most flattering to the headline Gamma share, so reporting it as the baseline would overstate the result. The sample itself supports the choice: regressing the VIX1D variance increments in the form $\Delta v / (\sqrt{v} \sqrt{\Delta t})$ gives a realized vol-of-vol of $\hat{\epsilon} = 0.90$, close to the chosen

baseline and nowhere near 0.5. The same exercise gives a realized price–variance correlation of $\hat{\rho} = -0.73$ against the assumed -0.7 , so of the two uncalibrated diffusion parameters ρ turns out to have been guessed well and ϵ badly.

Five-minute SPY data was used as a proxy for the underlying asset; the pipeline rescales each day to open at 1, so the results are invariant to the units of the price series. The sample covers 57 trading days, from May 11th, 2026 through July 31st, 2026. That window holds 60 weekdays, of which Memorial Day, Juneteenth, and the observed Independence Day holiday are closed, which is where the count of 57 comes from. The window is set by data availability rather than by choice: the intraday feed serves five-minute index data only for the trailing sixty trading days, so a longer or earlier sample is not obtainable from this source. The option prices were computed using the Heston model, and the hedging error was measured by rebalancing the delta hedge at each five-minute bar. The contributions of each Greek to the overall hedged profit and loss were then analyzed and their results are found in the following section. Better data would use actual option pricing instead of proxies, but option data is expensive and currently a limitation for this paper.

In terms of the derivation itself, the specific terms above were chosen to allow for all the terms to be expressed as Greeks, and have a discrete and nondiscrete component. The discrete component is the difference between the actual change in the underlying asset and the expected change, while the nondiscrete component is the expected change itself. The discrete component, following Bergomi’s 2016 book, is what contributes to the hedging error, while the nondiscrete component is what is expected to be hedged away. If the limit is taken, the model collapses back to the Heston model for every term except Vega: the other compensators go to zero with the hedging interval, while the Vega term converges to the unhedged variance risk described above.

In the implementation, the trapezoid rule was used to compute the integrals in the Heston pricer, and the Greeks were computed using finite differences. The code is available in the accompanying Python file on GitHub. The ω grid begins at 10^{-8} to avoid the $1/(i\omega)$ singularity at zero, and the upper limit of the integral is set to 2500 to ensure convergence for 0DTE options. The number of points in the grid is set to 2048, which is enough for the reported prices to stabilize. If using a stronger computer, the number of points can be increased to improve accuracy, but at the cost of longer computation time. With testing, it has been found that if λ is set to 0, then the results are not sensitive to the number of points in the grid, as the integrals

converge quickly. However, if λ is nonzero, then the results can be sensitive to the number of points in the grid, as the integrals converge more slowly. Also, setting λ to a nonzero value can introduce numerical instability in the integrals, as the integrands can become highly oscillatory, therefore making the pricer and the vega compensator disagree.

Two distinct experiments appear below, and the distinction governs how each residual should be read. The empirical study of Section 3.1 is not simulated at all: ΔS is the realized five-minute change in the observed underlying and Δv is the realized change in the squared VIX1D. The validation of Section 3 is simulated, using the Euler–Maruyama scheme to generate paths from the Heston dynamics.

On the simulated paths the price is carried in levels rather than log prices. An Euler step $\Delta S = rS \Delta t + \sqrt{v} S \Delta X$ has conditional variance $vS^2 \Delta t$, which is exactly the quantity the Gamma compensator $(\Delta S)^2 - vS^2 \Delta t$ subtracts. Because the paths are advanced in several sub-steps per hedge bar, simulating in levels keeps that compensator mean-zero up to an $O(\Delta t^2)$ correction, of the same order as the Taylor truncation itself, so any residual left in the *simulated* test reflects second-order error rather than a mismatch between how the paths were generated and how the error was decomposed. A log-Euler step would reintroduce such a mismatch through the $-\frac{1}{2}v$ Itô correction, which the level scheme does not carry.

No such guarantee is available on real data, though the gap narrows considerably once the variance proxy is horizon-matched. There ΔS is whatever the market did, while $vS^2 \Delta t$ is built from a VIX1D-derived variance that was never fitted to the realized path. Summed within each day, the realized quadratic variation $\sum (\Delta S)^2$ averages 4.75×10^{-5} across the sample, against 6.32×10^{-5} for the model’s $\sum vS^2 \Delta t$. The model quantity is about a third larger than the realized one, and it exceeds it on 51 of the 57 days. The Gamma compensator is therefore not mean-zero on the empirical sample but reliably negative, and because an at-the-money call is long Gamma the decomposition’s largest term carries a systematic negative drift. The residual reported in Section 3.1 accordingly mixes second-order Taylor truncation with error in the variance proxy, and the two residuals in this paper measure different things and should not be compared directly.

The variance path is read off Cboe’s VIX1D index, introduced in 2023 and computed from zero- to one-day SPY options [17]. The choice is deliberate and it matters. The thirty-day VIX is the wrong horizon for this experiment: VIX^2 is the expected *average* variance over the coming thirty days, whereas the model calls for the instantaneous spot variance v_t , and because variance mean-reverts a thirty-day forward average is pulled toward

the long-run level and barely moves within a session. Used as v_t it leaves the Vega, Vanna, and Volga terms only lightly exercised, and it overstates the variance an expiring contract actually faces. VIX1D is matched to the maturity the 0DTE option lives on and moves freely intraday, and the results below are built on it.

Two caveats attach to the substitution, one inherited and one new. The inherited one is that VIX1D² remains a risk-neutral quantity rather than a physical one, and it embeds a variance risk premium that the model, with $\lambda = 0$, does not capture. The new one is specific to the one-day horizon and is the more serious of the two. VIX1D measures expected variance over the coming twenty-four hours, so as a session advances its window rolls off the remainder of today and onto the overnight gap and the following session. For a contract expiring at today's close the remaining variance should decay toward zero over the session; VIX1D instead climbs, averaging 10.90 at 09:30 and 13.93 at 15:55 across the sample, rising on 55 of the 57 days at a Newey–West t of +8.5. The index is horizon-matched in name, but its intraday path carries a deterministic roll-up that the model reads as genuine variance news. Section 3.1 shows that this roll is what gives the Vega term its drift, and the drift should be read as a property of the proxy rather than of the market. Detrending the roll, or building a proxy that interpolates the term structure down to the actual remaining maturity, is the natural correction and is left to future work.

3 Validation & Results

According to Bertsimas, Kogan, and Lo (2000) [11], the granularity constant g , which measures how difficult an option is to hedge in discrete time and sets the size of the tracking error, blows up in the terminal interval. The Taylor expansion also fails as the option approaches expiry, which is one candidate explanation for the 20.2% gross residual reported in the next subsection.

The decomposition algebra itself was checked away from that corner. A 60-day option was simulated under the Heston engine and hedged daily, with the position closed 20 days before expiry so that no step straddles the payoff kink and every Greek stays smooth; there the seven terms reproduced the simulated hedged P&L to within 1.74%, at a correlation of 0.9998. That is a test of whether the expansion is correct, and it passes.

The terminal-interval explanation was then tested directly, and on the real data rather than by analogy to the 60-day run. The same 57 days were hedged the same way, but the position was closed a number of bars

before the bell and marked out at the pricer value instead of being held into expiry; every other element of the experiment is unchanged, so distance from expiry is the only thing that varies. The gross residual falls monotonically and steeply as the position is closed earlier: 20.2% held to expiry, 17.7% closed one five-minute bar early, 17.0% two bars early, 16.0% three bars early, 14.0% half an hour early, 10.5% one hour early, and 6.2% two hours early. Closing merely one bar before the bell removes two and a half points of residual, closing thirty minutes early removes nearly a third of it, and closing two hours early removes better than two thirds. The unexplained variance share falls along the same ladder, from 5.9% held to expiry to 1.0% two hours early. The residual is therefore concentrated in the final intervals, exactly as the granularity constant blowing up in the terminal interval would predict. This is a controlled version of the comparison the 60-day run could only gesture at: the maturity, hedge frequency, and data source are all held fixed, so the fall can be attributed to distance from expiry alone.

The 60-day simulation of the previous paragraph remains useful as an algebra check but is not the terminal-interval test: it differs from the 0DTE study in maturity, hedge frequency, and data source at once, and by the argument of Section 2.2 its residual is not even the same quantity, being second-order truncation on paths where the Gamma compensator is mean-zero to leading order, whereas the empirical residual also contains variance-proxy error. The direct truncation test above is the controlled statement; the 60-day run is corroborating context.

3.1 Results

The decomposition was tested on 57 full trading days of five-minute data (78 bars per day, so 77 hedge intervals). Each day is rescaled to open at 1, the variance path is read off the squared VIX1D, and the option is a fresh at-the-money 0DTE call re-formed each morning and delta-hedged at every bar. Summing the seven compensated terms reproduces the realized hedged P&L to within a 20.2% gross residual, defined as the mean absolute unexplained P&L across the 57 days divided by the mean absolute daily P&L. That metric is the harsher of the two reported here; measured against the *variance* of the realized hedged P&L, which is the decomposition the rest of this section leads with, the unexplained share is 5.9%. The mean daily hedged P&L is -0.00019 in the units where the underlying opens at 1, with a standard deviation across days of 0.00087.

That mean is not statistically distinguishable from zero, and the reason it is not is more informative than the mean itself. Daily P&L is serially

dependent on this sample, with a Ljung–Box $Q(10)$ of 25.9 ($p = 0.004$), so the independent-sampling standard error understates uncertainty and all inference below uses Newey–West errors at the automatic lag $L = 3$. With a standard deviation of 0.00087 across $D = 57$ days the Newey–West standard error of the mean is 0.00010, placing the average daily loss 1.9 standard errors below zero, which does not clear conventional significance; 40 of the 57 days lose money. The hedge does not lose persistently on this data.

Two large term-level drifts sit underneath that null, and they nearly cancel. The Gamma term averages -4.58×10^{-4} per day at a Newey–West t of -6.2 , negative on 49 of 57 days, and the Vega term averages $+3.77 \times 10^{-4}$ at $t = +6.5$, positive on 45. Neither is a risk premium. The Gamma drift is the variance input, measured in Section 2.2: the $vS^2 \Delta t$ subtracted inside the Gamma compensator runs about a third larger than the variance the underlying actually realized and exceeds it on 51 of the 57 days, and an at-the-money call is long Gamma, so the position pays that spread on most days in the sample. The Vega drift is the VIX1D roll-up of Section 2.2: the proxy climbs through the session on 55 of 57 days, the Vega term is $\partial V / \partial v [\Delta v - \nu_Q \Delta t]$, and a variance input that rises by construction hands that term a positive mean by construction. Both drifts are properties of the measurement rather than of the market, they point in opposite directions, and the near-zero net is the coincidence of two proxy errors rather than evidence that a 0DTE delta hedge breaks even. This sample cannot speak to the variance risk premium at all.

Table 1 reports each term two ways, and the two measures answer different questions. The *variance share* is a signed decomposition of the variance of the realized hedged P&L: term i ’s share is $\text{cov}(\text{term}_i, \Delta \Pi) / \text{var}(\Delta \Pi)$, so the seven terms and the unexplained remainder sum to 100% by construction and a term earns share only insofar as it moves *with* the error actually realized. This is the measure that answers where the hedging error goes, and it is the primary one here. The *gross share* is the model-attribution measure carried by earlier drafts of this work: each term’s average absolute daily contribution as a fraction of the sum across all seven, computed from model Greeks. It is weaker than the phrase “where the error goes” suggests, because it records how much a term *moves* and not how much of the realized error it accounts for; it is retained because it is the natural measure for the regime comparison, where the question is which Greek is being exercised rather than which is being realized.

The gap between the two measures is where a naive reading goes wrong, and Vanna shows it plainly: it takes 9.2% of the gross movement but a variance share of -1.5% with an interval spanning zero, so it moves without

contributing to the error that is realized. Across the 57 days the Gamma and Vega terms correlate at $\rho = +0.27$, with a 95% interval of $[-0.13, +0.58]$, so there is no evidence that the two terms carrying nearly all of the movement cancel within a day. The strongest offsetting pair anywhere in the matrix is Vega against Veta at $\rho = -0.71$, but Veta's daily standard deviation is smaller than Vega's by a factor of sixty-eight, so that cancellation is immaterial in P&L terms. In the table, Vega, Vanna, Volga, and Veta denote derivatives with respect to the variance v , not the volatility σ of Section 1; the two are related by chain rule but are not numerically identical. Both overall columns carry a percentile-bootstrap 95% interval over days ($B = 10,000$ resamples of the 57 days, shares recomputed on each). The regime columns split the sample into three groups by how far VIX1D travelled intraday, measured in index points.

Term	Variance share (%)		Gross share (%)		Gross by regime (%)		
	Share	95% CI	Share	95% CI	Calm	Mid	Turb.
Gamma ($\frac{1}{2}\partial^2 V/\partial S^2$)	64.6	[52.2, 73.9]	50.0	[45.4, 54.6]	62.9	46.9	44.9
Vega ($\partial V/\partial v$)	31.5	[17.5, 43.8]	34.7	[30.4, 39.3]	19.7	35.3	42.8
Vanna ($\partial^2 V/\partial S\partial v$)	-1.5	[-9.2, 7.0]	9.2	[7.4, 11.0]	10.1	10.5	7.7
Volga ($\frac{1}{2}\partial^2 V/\partial v^2$)	-1.8	[-2.9, -0.8]	2.0	[1.6, 2.4]	3.0	2.3	1.2
Charm ($\partial^2 V/\partial S\partial t$)	1.3	[0.2, 2.5]	1.8	[1.3, 2.4]	1.4	2.3	1.7
Time conv. ($\frac{1}{2}\partial^2 V/\partial t^2$)	0.3	[-0.4, 1.0]	1.6	[1.2, 2.0]	2.5	2.0	0.8
Veta ($\partial^2 V/\partial v\partial t$)	-0.3	[-0.5, -0.1]	0.7	[0.5, 0.8]	0.4	0.6	0.8
Unexplained	5.9	—	—	—	—	—	—

Table 1: Attribution of the hedged P&L across the 57 days, in percent. The variance share is the signed decomposition $\text{cov}(\text{term}, \Delta\Pi)/\text{var}(\Delta\Pi)$, so the seven terms plus the unexplained remainder sum to 100. The gross share is each term's average absolute daily contribution as a fraction of the summed average absolute contributions across all seven terms, and by construction admits no remainder. Intervals are percentile bootstraps over days ($B = 10,000$ resamples of the 57 days). The regime columns are gross shares within terciles sorted by the absolute intraday change in VIX1D, measured in index points: calm ($n = 19$, mean absolute change 0.90 points), mid ($n = 19$, 2.47 points), and turbulent ($n = 19$, 5.74 points). The variance shares within those same terciles are given in the text and are not monotone.

Gamma leads on both measures and in every regime, as one would expect of an option hedged only in the underlying, but it no longer dominates the way the thirty-day proxy made it appear: half the total absolute term

movement and two-thirds of the realized P&L variance sit in the $\frac{1}{2} \partial^2 V / \partial S^2$ term. Vega is the second term and it is not a minor one, taking 34.7% of the gross movement and 31.5% of the variance. A horizon-matched variance proxy moves enough within a session that the variance-facing part of the decomposition becomes a first-order object rather than a rounding error, and that is the single largest change from a thirty-day proxy.

What the regime changes is the balance between the two. Vega’s gross share rises from 19.7% on calm days to 42.8% on turbulent ones while Gamma’s falls from 62.9% to 44.9%, and in the turbulent tercile the two are within about two points of each other. The signed reading does not track that sort cleanly. Vega’s variance share goes 10.2%, 26.7%, 24.0% across the terciles and Gamma’s goes 77.2%, 65.5%, 73.9%: both move, neither is monotone, and with nineteen days per bucket the middle and turbulent figures are not separable. What does move monotonically is Vega’s correlation with the realized P&L, +0.27, +0.69, +0.71. Two claims carried by earlier drafts of this work do not survive the horizon-matched proxy and should be discarded. Vega’s dispersion does not grow faster than Gamma’s across the sort: its standard deviation across days grows by a factor of 1.4 against Gamma’s 1.5. And its mean is no longer indistinguishable from zero, running at Newey–West t of -0.4 , $+9.0$, and $+9.7$ across the three buckets. That drift is the roll-up of Section 2.2 rather than a market fact: the sort is on how far VIX1D travelled, VIX1D travels upward on 55 of 57 days, and so the turbulent bucket is disproportionately the bucket in which the proxy climbed most. Turbulence buys Vega a larger share of the movement; the accompanying drift is an artifact of how the variance was measured.

Vanna is the one term whose two measures disagree in sign, at 9.2% gross against -1.5% of variance with an interval spanning zero: it is exercised without contributing to the realized error. Charm, time convexity, Volga, and Veta stay within a few percent on both measures throughout, and by the bootstrap intervals in Table 1 the Charm and time-convexity shares are too close to rank against each other.

The regime pattern needs one caveat, and it is sharper here than it was under a thirty-day proxy. The Vega term is $\partial V / \partial v [\Delta v - \nu_Q \Delta t]$, so it is larger by construction on days when the variance proxy moved; sorting days by exactly that quantity and then observing that Vega’s share rises is partly circular. To check that the swing is a feature of turbulent markets rather than of the sort, the same 57 days were re-sorted into terciles by the absolute intraday move in the underlying, $|\Delta S|$, a criterion that does not appear in the Vega term. The swing survives: Gamma’s gross share runs 55.5% to 42.3% from the calmest to the most active tercile, and Vega’s 30.6% to

43.1%. The effect is attenuated relative to the $|\Delta\text{VIX1D}|$ sort but clearly present, so it is not an artifact of the sorting variable. The control is also a better one than it was: the two sorts correlate at 0.55 by Pearson and only 0.25 by Spearman on this sample, against 0.80 under the thirty-day proxy, so the underlying's move and the variance proxy's move are now close to independent orderings of the sample. The variance shares under the $|\Delta S|$ sort are again not monotone, at 71.4%, 39.4%, 71.4% for Gamma and 13.9%, 48.5%, 43.8% for Vega, which is a further reason to read the regime result as a statement about which Greek is exercised rather than about which carries the realized error.

Because Heston carries no jump term, any intraday jump enters the decomposition only through $(\Delta S)^2$ and is absorbed into the Gamma term, so the Gamma share reported above is a diffusion-plus-jump quantity. Separating the two inside the term requires a jump model and is beyond this decomposition, but the contamination can be bounded from the same five-minute bars. Following Barndorff-Nielsen and Shephard (2006) [13], realized variance $\sum r_i^2$ estimates total quadratic variation while bipower variation $\mu_1^{-2} \sum |r_i||r_{i-1}|$ is robust to finite-activity jumps, so their difference isolates the jump contribution; the ratio statistic is standardized by tri-power quarticity in the form given by Huang and Tauchen (2005) [14]. On the 77 five-minute returns of each session the test flags 12 of the 57 days as containing jumps at the 5% level and 3 at the 1% level, and the average relative jump contribution $\max(RV - BV, 0)/RV$ is 7.1%.

Re-running the attribution on the 45 jump-free days alone leaves the result intact. Gamma's gross share is 50.6% against 50.0% on the full sample, Vega's 34.4% against 34.7%, and the gross residual 19.9% against 20.2%; on the variance measure Gamma takes 59.3% against 64.6% and Vega 33.3% against 31.5%, with the unexplained share rising to 9.4%. Gamma's dominance is therefore not produced by the days on which jumps are detectable, and the headline survives on a clean subsample. Two limits on that conclusion should be stated. The subsample has 45 days rather than 57, so the intervals widen. More importantly the test discriminates weakly on this sample: the 12 flagged days on their own give a Gamma gross share of 47.8%, close to the jump-free figure, so the check bounds the contamination rather than demonstrating that jumps are irrelevant. What it rules out is the reading in which the Gamma share is an artifact of a handful of jump days.

The two remaining robustness questions are the mean-reversion rate κ and the vol-of-vol ϵ , both chosen rather than calibrated. Each was swept over three values with all other parameters and the data held fixed; Table 2

reports both. The two behave very differently. Varying κ across $\{1, 3, 5\}$ moves no share by as much as a third of a percentage point: the overall Gamma gross share runs 49.9%, 50.0%, 50.2% and the variance share is 64.6% in all three cases. This confirms numerically what the magnitude argument of Section 2.2 anticipated, that on a five-minute 0DTE horizon κ enters only through the negligible $\nu_Q \Delta t$ term and cannot affect the attribution. Varying ϵ across $\{0.5, 1.0, 2.0\}$ is a different story. Gamma remains the largest term throughout, but its gross share falls from 52.1% at $\epsilon = 0.5$ to 41.5% at $\epsilon = 2.0$ and its variance share from 69.1% to 55.3%. The gross share it sheds goes to Vanna and Volga, the cross- and second-variance terms that scale in ϵ . Vega's variance share is the one quantity in the table that ϵ leaves alone, sitting at 31.4%, 31.5%, and 30.8% across the sweep even as its gross share drifts down from 36.1% to 29.1%; the part of the realized error that Vega explains is a property of the data rather than of the vol-of-vol. The regime pattern of Table 1 holds in every row, with Gamma falling and Vega rising from the calm to the turbulent tercile. The gross residual falls slightly as ϵ grows, from 21.6% to 18.5%. The reading of the headline is that Gamma carries between two-fifths and one-half of the term movement and between five-ninths and seven-tenths of the realized variance, with the precise figure set by ϵ and not by κ , and that Vega is a stable second on the measure that matters.

	Gross share (%)				Var. share (%)		Turb. gross (%)	
	Gamma	Vega	Vanna	Volga	Gamma	Vega	Gamma	Vega
<i>Vary mean reversion κ (with $\epsilon = 1.0$):</i>								
$\kappa = 1$	49.9	34.8	9.2	2.0	64.6	31.6	44.8	42.9
$\kappa = 3$	50.0	34.7	9.2	2.0	64.6	31.5	44.9	42.8
$\kappa = 5$	50.2	34.7	9.1	2.0	64.7	31.5	45.0	42.7
<i>Vary vol-of-vol ϵ (with $\kappa = 3$):</i>								
$\epsilon = 0.5$	52.1	36.1	5.7	1.7	69.1	31.4	45.5	43.6
$\epsilon = 1.0$	50.0	34.7	9.2	2.0	64.6	31.5	44.9	42.8
$\epsilon = 2.0$	41.5	29.1	18.3	7.8	55.3	30.8	39.7	37.4

Table 2: Attribution shares as κ and ϵ are each swept, with the other parameters and the data held fixed. κ is inert: all three rows agree to within a third of a percentage point, as the magnitude argument of Section 2.2 predicts. ϵ matters for the gross measure, where Gamma cedes share to Vanna and Volga as it grows, but Vega’s variance share is stable across the sweep at 31%. The regime split holds in every row. The gross residual runs 21.6%, 20.2%, and 18.5% down the ϵ panel and is unchanged by κ . The $\kappa = 3$, $\epsilon = 1.0$ baseline is shared by both panels.

3.2 Limitations and Future Work

Four limits on the scope of these results should be stated plainly. The option studied is at the money at inception on every one of the 57 days, so the attribution is an at-the-money result and does not carry to the out-of-the-money strikes that make up much of 0DTE volume. The at-the-money case was chosen deliberately: it is where dealer gamma exposure concentrates [15], where the hedging problem is most acute, and where the Gamma-dominance result is most cleanly interpretable. Out-of-the-money strikes, which carry proportionally more Vega and Vanna, are left to future work. The parameters are a single uncalibrated set; the κ and ϵ sweeps of Section 3.1 show Gamma stays largest throughout but that its precise share depends on ϵ , and no sweep over η , ρ , or λ was attempted. The variance path is a one-day risk-neutral proxy standing in for an instantaneous physical variance: it runs about a third above the realized intraday variance, which flatters the Gamma share, and it carries the intraday roll-up documented in Section 2.2, which manufactures the Vega term’s drift. Neither distortion is removed here, and correcting the second is the most valuable next step this paper can name. Finally, the two attribution metrics measure different things and neither is complete on its own: the gross share records the size of

a term’s movement without regard to the realized error, while the variance share records covariance with the realized error and so is silent about terms that move a great deal without contributing to it.

The model is limited by the accuracy of the Heston parameters, an issue one can easily fix with better data. Another limitation is found in the premise itself: all Greeks are unhedgeable in the terminal interval, because there exists no rebalance prior to expiry. Given that, market makers and retail traders alike should be wary of 0DTE options, which are risky and difficult to hedge, especially as they near expiration. This paper locates the hedging error and does not provide a definitive solution to the problem. To make this model better, a Milstein scheme [4] could be used to simulate the variance process, which would provide a more accurate representation of the underlying asset’s volatility.

A further limitation is structural. Heston is a pure-diffusion model, so the decomposition carries no jump term; any intraday jump in the underlying enters only through $(\Delta S)^2$ and is therefore absorbed into the Gamma term, indistinguishable from ordinary diffusive movement. For 0DTE the omission is costly. Božović (2025) [16], working with five-minute SPY options, estimates the implied jump-risk premium at nearly twice the combined diffusion and volatility premia, with jump intensity clustering at the open and close. The Gamma share reported above should therefore be read as diffusion and jump risk combined; isolating the two would require adding a jump term to the decomposition. The bipower-variation check of Section 3.1 bounds how much this can matter, showing the attribution essentially unchanged on the subsample of days with no detectable jump activity, but bounding the contamination is not the same as decomposing it, and a jump-diffusion version of this exercise remains the cleaner answer.

4 Conclusion

Learning the different attributions of the error is important for understanding the risks of 0DTE options and using them effectively. Gamma carries the largest share on both measures reported here, two-thirds of the variance of the realized hedged P&L and half of the gross term movement, and the model conflates that share with jumps; a bipower-variation check on the jump-free days leaves it intact. Vega is a clear second rather than a rounding error, taking about a third on both measures. It is the variance risk a delta-only hedge cannot remove at any rebalancing frequency, and its gross share more than doubles from the calmest to the most turbulent days, a pattern that

survives a re-sort on the underlying's own move rather than the variance proxy's. The other Greeks, while not negligible, are relatively small; Vanna is the instructive case, moving a great deal without contributing to the error that is realized. These results are preliminary. They describe at-the-money calls at a single baseline parameter set, robust to the mean-reversion rate but not to the vol-of-vol, with a one-day variance proxy that runs about a third above realized variance and that climbs mechanically through the session. That last defect is the binding one: it manufactures the drift in the Vega term, and until it is corrected this sample can say nothing about whether a 0DTE delta hedge loses money on average. Further work is needed before the attribution can be claimed for 0DTE hedging in general. Future work should focus on correcting the intraday roll in the variance proxy, improving the Heston parameter estimates, adding a jump term to the decomposition, and testing the model on a wider range of data.

A Glossary of Variables

The sensitivity Greeks (Δ , Γ , ν , Θ , and the higher-order terms) are defined in Section 1. The symbols below collect the model, market, and hedging notation that the Heston and derivation sections introduce along the way.

Underlying and market.

- S : price of the underlying asset (the spot), rescaled so that each trading day opens at 1.
- v : instantaneous variance of the underlying; the volatility is $\sqrt{v}$.
- σ : volatility, with $\sigma = \sqrt{v}$; used in the Greek definitions of Section 1.
- t : calendar time. Written as a prefix, Δt is a single hedging interval (one five-minute bar in the test).
- μ : real-world (physical) drift of the underlying.
- r : risk-free interest rate ($r = 0.045$ in the test).
- X, W : the Brownian motions driving the price and the variance, correlated by $dX dW = \rho dt$.

Heston parameters.

- κ : rate at which the variance reverts toward its long-run level.

- η : long-run (mean) level of the variance.
- ϵ : volatility of volatility.
- ρ : correlation between the price and variance shocks.
- λ : market price of volatility risk; $\lambda = 0$ in the test, so the physical and risk-neutral variance drifts coincide.
- ν_Q : risk-neutral drift of the variance, $\nu_Q = \kappa(\eta - v) - \lambda v$ (distinct from Vega ν).

Option and hedge portfolio.

- V : value of the option being hedged (a fresh at-the-money 0DTE call).
- V_1 : value of the second option held to hedge variance (Vega) risk.
- Π : value of the delta–vega hedged portfolio, $\Pi = V - \Delta S - \Delta_1 V_1$.
- Δ : units of the underlying held in the hedge, $\Delta = \partial V / \partial S - \Delta_1 \partial V_1 / \partial S$. Written as a prefix (ΔS , Δv , $\Delta \Pi$) it instead denotes the change in a quantity over one hedging interval.
- Δ_1 : units of the hedging option V_1 held, chosen to cancel the portfolio’s net Vega, $\Delta_1 = (\partial V / \partial v) / (\partial V_1 / \partial v)$.

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